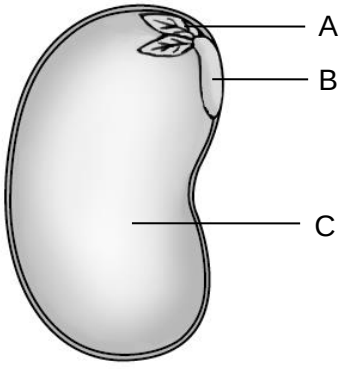
Short questions

|!|BCESC0150000001|!|
The diagram below shows the longitudinal section of a seed.



(a) What would structures A and B develop into after seed germination? (2 marks)

(b) Amylases are found in structure C during seed germination. Suggest the significance of the presence of amylases in structure C for the growth of the seedling. (2 marks)

(c) The dry mass of the seedling decreases in the first few days of germination. The dry mass starts to increase afterwards. Explain the change of the dry mass of the seedling. (3 marks)

##

(a) Structure A would develop into a shoot. (1)
Structure B would develop into a root. (1)

(b) Amylases break down the carbohydrates in structure C into simple sugar, (1)

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which can be used for respiration / cell formation / growth. (1)

(c) The seedling uses the food reserves in structure C for growth, therefore the dry mass of the seedling decreases in the first few days. (1)

After a few days, the seedling develops leaves and is able to carry out photosynthesis. (1)

When the rate of photosynthesis is higher than the rate of respiration / rate of food consumption, there is net gain in dry mass. (1)

(7 marks)

##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	3

|!|BCESC0150000002|!|

Complete the following paragraph with suitable words. (5 marks)

When a seed germinates, (a)_____ break down the starch stored in the (b)_____ into glucose, proteins into amino acids, and lipids into fatty acids and (c) _____. These food substances are used in (d)_____ to release energy and are transported to the growing regions of the embryo to synthesize materials for new cells. As a result, the radicle develops into a root, and the (e)_____ develops into a stem and leaves.

##

(a) enzymes (1)

(b) cotyledons (1)

(c) glycerol (1)

(d) respiration (1)

(e) plumule (1)

(5 marks)

##

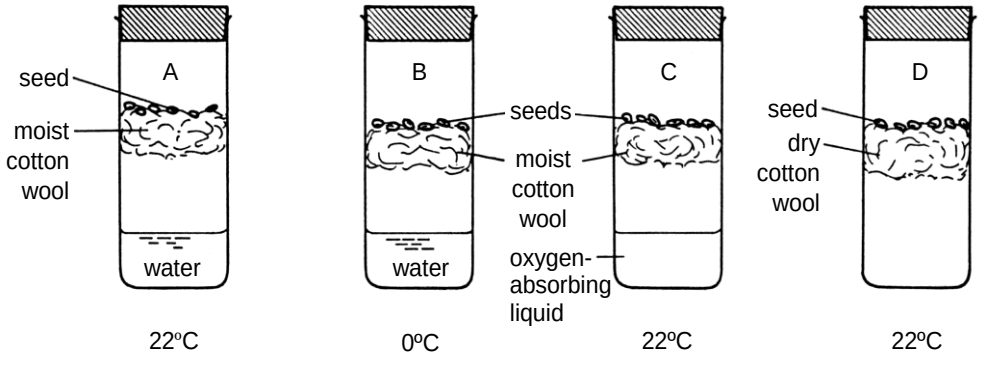
	Skill assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1

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(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESB0150000003|!| ★ **SI**

The following diagram shows an experiment that is used to study the conditions necessary for seed germination.



(a) If only those seeds in tube A germinate, what would be the conditions necessary for seed germination? (3 marks)

(b) How would you determine whether light is necessary for seed germination? (2 marks)

##

(a) Oxygen (1)
Water (1)
Warmth / optimum temperature (1)

(b) Set up a tube (tube E) similar to tube A, but covered by a piece of black paper / aluminium foil. (1)
If seeds in tube E do not germinate, we can conclude that light is required for seed germination. (1)

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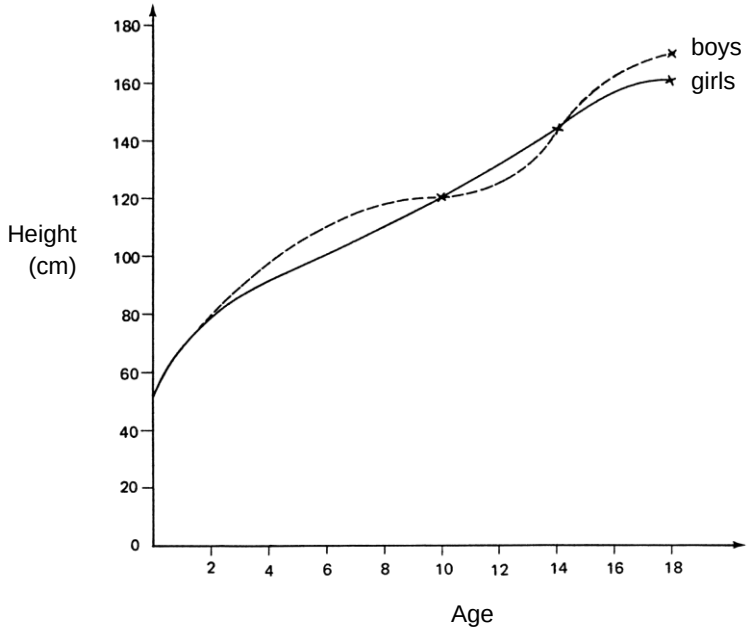
(5 marks)

##

	Skill assessed	Marks
(a)	Applying concepts	3
(b)	Practical skills	2

|!|BCESC0150000004|!|

The graph below shows the changes in the mean heights of thousands of boys and girls with age.



(a) In which age range(s) is the growth of girls faster than that in boys? (1 mark)

(b) In which age range(s) is the growth of boys faster than that in girls? (1 mark)

(c) State ***two*** factors that affect the height gain in a person. (2 marks)

##

(a) Between age 10 and 14 (1)

(b) Between age 2 and 10; over age 14 (1)

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(c) Any ***two*** of the following: (1 mark each) (2)

- diet
- heredity
- exercise

(Or other reasonable answers)

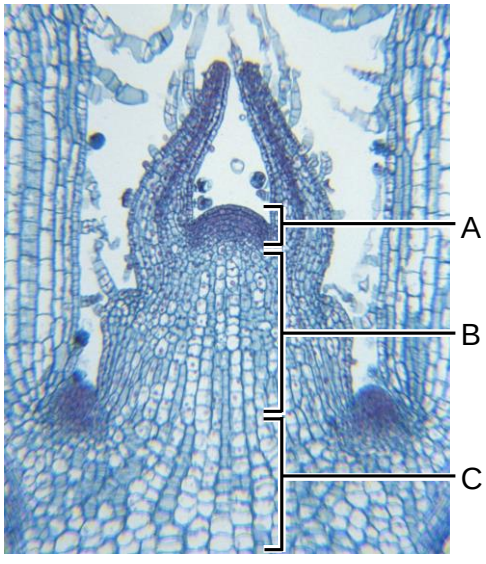
(4 marks)

##

	Skill assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Understanding basic concepts	2

|!|BCESC0150000005|!| **Cross topic**

The following photomicrograph shows a shoot tip.



(a) Identify regions A, B and C. (3 marks)

(b) Describe the role of region A in the growth of the plant. (1 mark)

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(c) Describe how the shoot growth is affected by the plant hormone, auxin.(2 marks)

##

(a) A: region of cell division (1)
B: region of elongation (1)
C: region of differentiation (1)

(b) Cells in region A divide actively by mitotic cell division, producing many small, thin-walled cells. (1)

(c) Auxins are produced at the shoot tip and diffuse to the region of elongation. (1)
A high concentration of auxins promotes the growth at the region of elongation. (1)

(6 marks)

##

	Skill assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2

|!|BCESC0150000006|!|

(a) What is the difference between fresh mass and dry mass of an organism?
(1 mark)

(b) Describe and explain the changes in the dry mass of a plant when it develops from a seed into a leafy plant. (5 marks)

##

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(a) Fresh mass includes the water content of the organism while dry mass does not include. (1)

(b) Since the food reserves in the seed are broken down to provide energy and raw materials for growth, (1)

the dry mass of the seed decreases at first. (1)

Once green leaves appear above the ground, photosynthesis takes place. (1)

When the rate of photosynthesis is higher than the rate of respiration, the production of food is faster than the consumption of food. (1)

This results in a net gain in the amount of organic materials. Dry mass increases. (1)

(6 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	5

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